

Topic: Recursive Queries

Q. What are Recursive Queries in SQL? Explain the concept of Common Table Expressions (CTE) and how a Recursive CTE works with an example.

****Definition to Remember****

A **Rec
hierarch
`WITH R

Recursive Queries & Common Table Expressions (CTE)

A CTE is a temporary, named result set defined using the `WITH` keyword. It exists only during the execution of a single SQL statement.

* Used

2. Structure of a Recursive CTE

A recursive CTE has exactly 3 components:

WITH RECURSIVE CTE_Name AS (

ANCHOR MEMBER <- Base case (executes once, gives R)

SELE

UNION ALL <- Connects anchor + recursive

RECURSIVE MEMBER <- References CTE itself (gives R, R, ...)

SELE

)

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3. Working of a Recursive Query

1. Anchor Member executes first -> produces base result set R_0.

2. Recu

4. Example - Organizational Hierarchy

Table: `Employee (EmpID, Name, ManagerID)` - Find all levels of management.

sql

UNION ALL

-- Recursive: Find employees managed by those already found

5. Applications

* **Organizational Charts:** Employee hierarchy traversal.

* **Bill**

Must-Write Points (for 10 marks)

1. Recu

Quick Recall

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